

Rylee Albrecht

Orlando, FL

(425) 283-8821 • rylee@albrecht.us • rylee.albrecht.us

EDUCATION

University of Central Florida (UCF) - M.S. Computer Vision, GPA: 3.97	Dec 2026
Brigham Young University (BYU) - B.S. Computer Science, Data Science Emphasis, GPA: 3.67	Jun 2024

SKILLS

- **Programming Languages:** Python, Java, JavaScript, SQL, R, C, C++, C#
- **ML/AI/Data Science:** PyTorch, transformers, Hugging Face, scikit-learn, OpenCV, CUDA, NumPy, pandas, matplotlib
- **Tools/Technologies:** Git, Jupyter, Docker, Bash, AWS, LaTeX, Unity, MATLAB

AWARDS

1st Place, Data Science - BYU Student Research Conference	Feb 2024
<i>Research: Mimicking Persuasive Conversational Dynamics with Language Models</i>	

EXPERIENCE

Center for Research in Computer Vision (UCF)	Mar 2026 – Present
<i>Research Assistant</i>	Orlando, FL

- Contributing to **FlexGait**, a self-training gait recognition framework for limited-label learning under UDA and SSL.
- Evaluating cross-domain generalization across multiple datasets and input modalities, including silhouette and pose.
- Conducting experiments and preparing the work for submission to CVPR 2027.

Photography Agent	Sept 2025 – Dec
<i>Collaborative project</i>	Orlando, FL

- Fine-tuned Llama-3-8B-Instruct with LoRA on ~2K self-curated examples to create an LLM-based photo editing agent.
- Contributed to the design and integration of tool-selection logic for invoking photo editing and filtering tools.
- Evaluated agent on test set, achieving 88% accuracy in tool selection, error handling, and conversational responses.

Harvard Aging Brain Study (Massachusetts General Hospital)	May 2024 – Aug
<i>Research Intern</i>	Boston, MA

- Preprocessed NIfTI neuroimages, including normalization and registration to MNI space.
- Worked on adapting a diffusion model for cross-modality MRI synthesis (T1-T2) to reduce the need for additional scans.
- Performed statistical analysis and visualization in MATLAB and R on neurological datasets.

Perception, Control, and Cognition Lab (BYU)	May 2023 – May 2024
<i>Research Assistant</i>	Provo, UT

- Engineered prompts across LLMs (e.g., LLaMA-2, GPT-4, Claude) to optimize and evaluate human-like responses.
- Fine-tuned large-scale LLaMA models on an HPC cluster using SLURM for distributed training.
- Tested LLM prompts at small scale to reduce costs before full-scale experiments.

Computational Health Sciences Lab (BYU)	Nov 2023 – Apr 2024
<i>Research Assistant</i>	Provo, UT

- Cleaned, aggregated, and engineered features from ER data to prepare structured inputs for modeling.
- Trained ML models and analyzed feature importance to examine potential contributing factors to ER diagnoses.
- Collaborated in an interdisciplinary lab across Statistics, Computer Science, and Public Health.

Bean Lab (BYU)	Sept 2023 – Apr 2024
<i>Capstone Data Science project</i>	Provo, UT

- Implemented cross-lingual semantic search using OpenAI embeddings and cosine similarity for passage retrieval.
- Integrated embedding pipelines into a full-stack application and validated system performance.

VOLUNTEER EXPERIENCE

The Church of Jesus Christ of Latter-Day Saints	Mar 2020 – Aug 2021
<i>Volunteer Full-Time Representative</i>	Richmond, VA

- Trained and mentored peers, coordinated team efforts, and worked with individuals from diverse backgrounds.